

Hard Work beats talent when talent
doesn't work hard,

— Tim Nothe.

$$16 = 2^{(4)} \quad \begin{array}{c} \text{base} \nearrow \quad \nwarrow \text{exponents} \\ \end{array} \quad 8 = 2^{(3)}$$

$$16 * 8 = 128$$

$$\begin{aligned} 16 * 8 &= 2^4 * 2^3 \\ &= 2^7 \\ &= 128 \end{aligned}$$

$$\left. \begin{array}{l} 16 \longrightarrow 4 \\ 8 \longrightarrow 3 \end{array} \right\} 7 \text{ bits} \Rightarrow 2^7 = 128$$

Multiplication $\longrightarrow$ Addition

$$2^x = y \quad \equiv \quad x = \log_2 y$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32 \downarrow$$

$$\log_2 2 = 1$$

$$\log_2 4 = 2$$

$$\log_2 8 = 3$$

$$\log_2 16 = 4$$

$$\log_2 32 = 5 \downarrow$$

$$3^2 = 9$$

$$\log_3 9 = 2$$

$$5^3 = 125$$

$$a) \log_2 125 = 5$$

$$b) \log_3 125 = 5$$

$$\checkmark c) \log_5 125 = 3$$

$$d) \log_2 125 = 3^5$$

$$\log_5 125$$

$$\log_2 32 = 5$$

$$\log_3 32 = 3.155$$

$$32 = 3^{3.1546\dots}$$

$$2^5 = 32 = 3^{3.155}$$

[bigger base $\rightarrow$ lesser exponent]

$$\log_2 1 = 0 \quad \log_3 1 = 0$$

$$\log_{10} 10000 = 4 \Leftrightarrow 10^4 = 10000$$

$$\log_b 1 = 0$$

$$\log_3 243 = 5$$

$$\log_9 243 \in [2, 3]$$

2.5

$$9^1 = 9$$

$$9^2 = 81$$

$$9^3 = 729$$

$$243 = 9^{? \text{ (exponent)}}$$

(val) (base)

$$9^x = 243$$

$$\Rightarrow (3^2)^x = 3^5$$

$$\Rightarrow 3^{2x} = 3^5$$

$$\Rightarrow 2x = 5$$

$$\Rightarrow x = 2.5$$

$$\underbrace{3^2 * 3^2 * 3^2 * \dots * 3^2}_x$$

$$3^? = 32$$

$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 27$$

$$3^4 = 81$$

$$3^x = 32$$

$$x \in [3, 4]$$

1st Law

$$x = a^n$$

$$y = a^m$$

$$\log_a x = n$$

$$\log_a y = m$$

$$xy = a^n * a^m = a^{n+m}$$

$$\Rightarrow \log_a xy = \log_a (a^{n+m}) = n+m$$

$$= \log_a x + \log_a y$$

$$\log_a xy = \log_a x + \log_a y$$

$$\log_7 30 = \log_7 (2 * 3 * 5)$$

$$= \log_7 2 + \log_7 3 + \log_7 5$$

2nd Law

$$x = a^n$$

$$\longrightarrow \log_a x = n$$

$$x^m = (a^n)^m = a^{nm}$$

$$\Rightarrow \log_a x^m = \log_a (a^{nm}) = nm = (\log_a x) * m$$

$$\log_a x^m = m \log_a x$$

3rd Law

$$\log_a (x/y) = \log_a x - \log_a y$$

$$\log_5 25 = 2$$

$$\log_{25} 5 = 1/2$$

$$\left[\log_b a = \frac{1}{\log_a b} \right]$$

$\log_a x \Rightarrow$ What can be the value of a ?

$$a \in (0, \infty) - \{1\}$$

$$x \in (0, +\infty)$$

$$\boxed{\log_1 2 = \cancel{x}}$$

$$\log_{0.5} 2 = -1$$

$$(0.5)^y = 2$$

$$\Rightarrow \left(\frac{1}{2}\right)^y = 2 \Rightarrow y = -1$$

$$3^x = 5^{x-2}$$

$$\Rightarrow \log_2(3^x) = \log_2(5^{x-2})$$

$$\Rightarrow x \log_2 3 = (x-2) \log_2 5$$

$$\Rightarrow x \log_2 3 = x \log_2 5 - 2 \log_2 5$$

$$\Rightarrow x (\log_2 5 - \log_2 3) = 2 \log_2 5$$

$$\Rightarrow x \log_2 \left(\frac{5}{3}\right) = 2 \log_2 5$$

$$\Rightarrow x = 2 * \frac{\log_2 5}{\log_2 (5/3)}$$

$$x = 2 * \frac{\log_b 5}{\log_b (5/3)}$$

$$b \in (0, \infty) - \{1\}$$

Common Base for Logarithm = 10.

$$\log_{10} 10^6 = 6 \longrightarrow 7 \text{ digits}$$

$$\log_{10} 98432 = 4.99 \dots$$

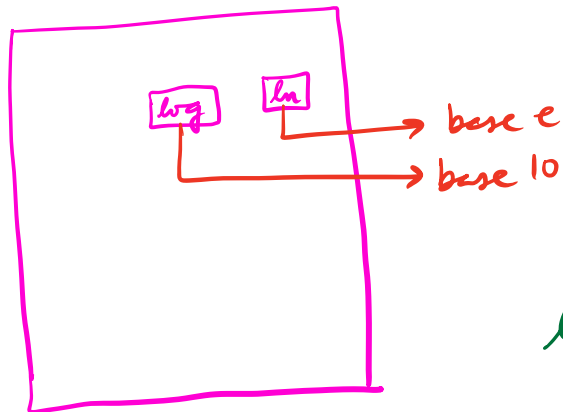
$$\lfloor \log_{10} 98432 \rfloor = 4 \longrightarrow 5 \text{ digits}$$

$$\lfloor \log_{10} 98 \rfloor = 1 \longrightarrow 2 \text{ digits}$$

$$\lfloor \log_{10} 9 \rfloor = 0 \longrightarrow 1 \text{ digit}$$

$$\lfloor \log_{10} n \rfloor + 1 = \text{no. of digits in } n$$

Natural Base for Logarithm $\rightarrow e = 2.7182818\dots$
 [Irrational number]



$$\log 25 \downarrow \log_{10} 25$$

$$\ln 25 \downarrow \log_e 25$$

Math. $\log(25) \Rightarrow \log_e 25$
 Math. $\log_{10}(25) \Rightarrow \log_{10} 25$

[Break till 10:51 PM]

$$\log_b a = \frac{\log_e a}{\log_e b} = \frac{\log_{10} a}{\log_{10} b} = \frac{\log a}{\log b}$$

$$\log_4 64 = \frac{\log_2 64}{\log_2 4} = \frac{6}{2} = 3$$

$$\text{If no. of iterations} = \log_2 n = \frac{\log_e n}{\log_e 2} = \frac{\log_3 n}{\log_3 2}$$

$$\log_2 n = \frac{\log_e n}{\log_e 2} = \left(\frac{1}{\log_e 2}\right) * \log_e n$$

$$= c * \log_e n$$

$$\log_3 n = \frac{1}{\log_e 3} * \log_e n$$

$$= \frac{1}{\log_2 3} * \log_2 n$$

$$\begin{array}{l} \log_3 n \rightarrow \\ \log_2 n \rightarrow \\ \log_a n \rightarrow \end{array} O(\log n)$$

$$128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

$$n = 2^{30}$$

$$\left. \begin{array}{l} \downarrow /2 \\ \downarrow /2 \\ \vdots \\ 1 \end{array} \right\} 30 \text{ steps}$$

$$\left. \begin{array}{l} \downarrow /3 \\ \downarrow /3 \\ \vdots \\ 1 \end{array} \right\} 30 * \log_3 2$$

$$n = 2^{30}$$

$$\downarrow$$

$$2^{29}$$

$$\downarrow$$

$$2^{28}$$

$$\downarrow$$

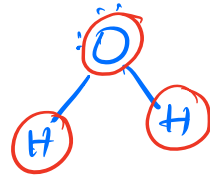
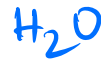
$$\vdots$$

$$\downarrow$$

$$2^0 = 1$$

$$\left(\begin{array}{l} 30 \\ \text{iterations} \end{array} \right) \rightarrow \log_2 n$$

iterations $\equiv$ molecules
 10^9 instructions $\equiv$ atoms



5 molecules of $\text{H}_2\text{O} \rightarrow 3 * 5 = 15$ atoms

15 atoms $\rightarrow$ 5 molecules



1 molecule of $\text{N}_2\text{O}_3 = 5$ atoms

15 atoms $\rightarrow$ 3 molecules.

```
long t1 = System.currentTimeMillis();
```

```
for (i=1; i <= 108; i++) {
```

```
    // ...
```

```
}
```

```
long t2 = System.currentTimeMillis();
```

```
System.out.println((t2-t1)*1.0/1000000);
```

time elapsed.

```

fn(i=0; i<n; i+=k) {
    fn(j=i+1; j<=k; j++) {
        // code
    }
}

```

$n=10$
 $k=4$

$i=0, j=1$
 $i=0, j=2$
 $i=0, j=3$
 $i=0, j=4$
 $i=4, j=5$
 $i=8, j=9$

$O(n)$.

$\frac{n}{k} \left\{ \begin{array}{l} i=0 \rightarrow j \in [1, k] \rightarrow k \\ i=k \rightarrow j \in [k+1, k] \rightarrow \phi \\ i=2k \rightarrow \phi \\ \vdots \rightarrow \phi \\ i < n \end{array} \right.$

$$k + \frac{n}{k}$$

$$k \in [1, n]$$

$$\frac{d(k + \frac{n}{k})}{dk} = 1 - \frac{n}{k^2} = 0$$

$$\Rightarrow \frac{n}{k^2} = 1 \Rightarrow k^2 = n$$

$$\Rightarrow k = \pm \sqrt{n}$$

$$\left. \begin{array}{l} k=1 \Rightarrow 1 + \frac{n}{1} = n+1 \rightarrow O(n) \\ k=n \Rightarrow n + \frac{n}{n} = n+1 \rightarrow O(n) \\ k=\frac{n}{2} = \frac{n}{2} + \frac{n}{n/2} = \frac{n}{2} + 2 \rightarrow O(n) \\ k=\sqrt{n} = \sqrt{n} + \frac{n}{\sqrt{n}} = 2\sqrt{n} \rightarrow O(\sqrt{n}) \end{array} \right\}$$

```

int i=1;
while (i < n) {
    int x=i;
    while (x-- > 0) {
        // O(1) op.
    }
    i++;
}

```

	<u>inner loop</u>
i=1	1
i=2	2
i=3	3
i=4	4
⋮	
i=n-1	n-1
	<u>1+...+n-1</u>

$$1 \dots n \rightarrow \frac{n(n+1)}{2}$$

$$1 \dots (n-1) \rightarrow \frac{(n-1)((n-1)+1)}{2}$$

$$= \frac{(n-1) * n}{2} = \frac{n^2}{2} - \frac{n}{2} = O(n^2)$$

```

i=n
while (i > 0) {
    i /= 2;
}

```

```

fn(i=0; i < n; i++) {
    fn(j=0; j < n; j++) {
        // O(1) op.
        break;
    }
}

```

i=0	j=0	1
i=1	j=0	1
i=2	j=0	1
i=3	j=0	1
⋮		
i=n-1	j=0	1
		<u>n</u>